

Clinical-row validation — human trials, August 2026

STATUS: DATED RECORD, PARTLY SUPERSEDED — read with `ROCKSTEADY_RESPONSE_KIDNEY.md`.

This file records the August-2026 validation round that first identified the provenance/outcome confound. Three of its statements have since been overtaken and are retained here as history rather than as current guidance:

(1) the `renal_endpoints_measured` field it recommends is now implemented on all 246 rows, joined by `negative_eligibility` and `nephrotox_grade_modeling`;

*(2) its "weakened 3.7×" / risk-difference framing is **withdrawn** — see `scripts/confound_stats.py`, which finds no measurable weakening;*

(3) its blanket "should not be used to train a nephrotoxicity model" is superseded by the per-row eligibility gate, which blocks the 20 specific rows rather than the dataset.

Its §2 per-row retrieval findings stand and are still the basis for several reclassifications.

Validation of the 39 `study_type=clinical` rows against their cited primary sources, prompted by a structural concern found before any source was opened (§1). Seven of the 13 unverified absence claims were checked directly; results in §2. **Grades in `measurements.csv` were NOT changed** — recommended reclassifications are recorded here for review, because relabelling a toxicity dataset is a scientific call, not a curation one.

1. The confound that motivated this

Cross-tabulating provenance against outcome across all 39 clinical rows:

	grade ≥2	grade <2	total
WS (search-derived)	0	20	20
anchor-sourced	11	8	19

- Base rate of grade ≥2 = 11/39 = 28.2%; expected among 20 unverified = **5.6**, observed **0**.
- One-sided Fisher exact **p = 4.5 × 10^{−10}**.
- 13 of 14 rows whose entire claim is an absence statement (`no_signal`/`none`/`no_change`) are WS.

Verification status predicts the label almost perfectly. A model trained on this as-is partly learns "*does this compound have a regulatory paper trail*" rather than renal biology — and that shortcut inverts on prospective compounds, which have no dossier and would be scored non-nephrotoxic by construction. That is the wrong error direction for a safety model.

2. What direct source retrieval found

The decisive question per row: does "no signal" mean (a) renal endpoints were **measured and unremarkable**, (b) renal endpoints were **never measured**, or (c) the **cited source does not report them**? Only (a) supports a grade of 0.

mid	drug	verdict	what the source actually shows
MSR066	cemdisiran	CONFIRMED	eGFR measured: change at wk32 −2.9 (11.1) vs placebo −6.3 (4.8) mL/min/1.73 m ² . A genuine measured negative, in fact favourable vs placebo.

mid	drug	verdict	what the source actually shows
MSR045	lumasiran	PARTIAL	eGFR reported extensively but as <i>efficacy/eligibility stratification</i> , not toxicity. AE table (Table 2) lists only injection-site reaction and abdominal pain. Renal function tracked; renal safety not assessed.
MSR063	pelacarsen	REFUTED	Source states "no excess nephrotoxicity" as a generic claim with no renal endpoint behind it — and the trials excluded eGFR <60 and UACR >100 mg/g. The design removes the population where nephrotoxicity would appear.
MSR078	fazirsiran	REFUTED	Safety assessments were "TEAEs and changes in FEV1 and DLCO Hb" only. No renal endpoint. Patients with renal dysfunction excluded at enrollment.
MSR042	tofersen	REFUTED	Label has no Renal Impairment subsection and no renal endpoint in the AE table. Sole renal mention: "No clinical studies have been conducted to evaluate the pharmacokinetics of tofersen in patients with renal or hepatic impairment." Intrathecal route plausibly explains low renal risk, but that is a mechanistic inference, not a measurement.
MSR044	patisiran	UNSUPPORTED	Onpattro label carries no renal endpoint in the AE table; only \$8.7 boilerplate dosing and "has not been studied in severe renal impairment or ESRD".
MSR047	vutrisiran	UNSUPPORTED	Amvuttra label identical in kind: AE table lists pain in extremity, arthralgia, dyspnea, vitamin A decreased. No renal endpoint.

Score: 1 of 7 checked absence claims survives as a measured negative.

The mechanism

Whether renal endpoints were measured tracks the **indication**, not the drug's actual renal risk. Cemdisiran (IgA nephropathy) and lumasiran (primary hyperoxaluria) have renal indications and renal data. Pelacarsen, fazirsiran, tofersen, patisiran and vutrisiran do not, and their trials neither measured renal endpoints nor enrolled renally impaired patients. The negative class is therefore substantially "*nobody looked*" rather than "*looked and found nothing*" — exactly as the confound predicted.

A second, subtler point for the approved drugs: absence of a renal AE from a label's adverse-reactions table is **not** a measured negative. AE tables list events above a frequency threshold, so a real low-frequency renal signal would not appear. Labels cannot settle these rows; the FDA Pharmacology/Toxicology reviews or the primary trial papers can.

3. Recommended reclassification (NOT applied)

Introduce a `renal_endpoints_measured` field with values `measured_and_reported` / `not_measured` / `not_reported_in_source` / `cannot_determine`, and set `nephrotox_grade` = `not_assessed` wherever it is `not_measured_and_reported`.

- MSR042, MSR063, MSR078 → `not_assessed` (renal endpoints never measured)
- MSR044, MSR047 → `not_assessed` (not reported in the cited source; resolvable via reviews/trial papers)
- MSR045 → keep 0, annotate as efficacy-derived rather than safety-assessed
- MSR066 → keep 0, and replace `no_signal` with the quantitative value (eGFR -2.9 mL/min/1.73 m²)

The six remaining unchecked absence claims (MSR049 pegaptanib, MSR055 fitusiran, MSR056 mongsersen, MSR064 zilebesiran, MSR067 donidalorsen, MSR079 vupanorsen) should be presumed suspect until retrieved; four of them are behind sources this environment cannot reach.

Until this field exists, the dataset should not be used to train a nephrotoxicity model. That recommendation rests on the aggregate counts in §1 and the 7 retrievals in §2.

4. Method note

Two attempted multi-agent validation runs returned nothing: every parameterized subagent tool call was rejected by a harness-level permission fault (`updatedInput` stripped of required parameters; only parameterless tools such as `ListAgents` succeeded). ~2.4M tokens produced zero retrievals across the two runs. All findings above were obtained by direct inline retrieval instead, which worked without incident.

5. Update — 3 September 2026

Three approved DMD PMOs (golodirsen, casimersen, viltolarsen) were added from their DailyMed labels, each contributing a human grade-0 row of the **measured** kind: the labels prescribe the analytes (serum cystatin C, urine dipstick, UPCR — monthly and quarterly) and then state affirmatively that "kidney toxicity was not observed in the clinical studies". Endpoints prescribed, result negative — the distinction §2 found missing from the WS rows.

Effect on the confound: anchor-sourced grade-0 clinical rows **1 → 4**, and the association weakens from one-sided Fisher **$p = 4.5 \times 10^{-10}$ to $p = 1.65 \times 10^{-10}$** . That is **No measurable weakening is claimed**. An earlier version of this analysis reported that adding three label-derived rows "weakened" the confound. That claim is withdrawn: the unverified arm is unchanged at **0 of 20** reaching grade ≥ 2 before and after, so every bit of the apparent movement came from three rows entering the *other* arm's denominator — and this release reclassifies those same three rows (MSR160/162/164) as `asserted_negative_regulatory` and gates them out of modelling. A weakening resting on rows the release declares unfit is not a finding. Computed by `scripts/confound_stats.py`: on all clinical rows the risk difference is **50.0 pp** (one-sided Fisher $p = 1.65 \times 10^{-10}$); on the rows that pass `negative_eligibility` it is **61.1 pp** ($p = 0.045$) — *larger*, on far less data, because the unverified arm collapses to 4 rows.

These three also strengthen the extrapolation evidence in the right direction. All three are now bridge compounds with animal grade 2 against human grade 0, and unlike lumasiran and vutrisiran their human negatives are supported — so of the 6 current `animal_over_predicts` verdicts, 3 now rest on measured human negatives rather than unexamined ones.